Theory of Interest

Accumulation Function Simple interest: $a(t) = (1+rt)$, Compound interest: $a(t) = (1+r)^t$. $r^{(p)}$ = Nominal yield compounded p times a year, interest in one period = $\frac{r^{(p)}}{p}$, interest per year = $(1+\frac{r^{(p)}}{p})^p$

Effective Interest On the interval $[t_1, t_2]$ = $\frac{a(t_2)-a(t_1)}{a(t_1)}$

Continuously Compounded Rate $a(t) = e^{rt}$. $e^r > (1+\frac{r}{p})^p \geq 1+r$ for $r > 0$ and $p \geq 1$

Equivalent IR Two rates are equivalent iff they yield the same accumulation amount over a year, $(1+\frac{r^{(p)}}{p})^p = (1+\frac{r^{(q)}}{q})^q$

Effective annual interest rate $1+r_e = (1+\frac{r^{(p)}}{p})^p$.

For any positive integer p, $r_e \equiv (1+\frac{r^{(p)}}{p})^p - 1 \geq r^{(p)}$, $a(t) = (1+r_e)^t$

Present Value (discounting) For $C = \{(c_1, t_1), \ldots, (c_n, t_n)\}$, $PV(C) = \sum_{i=1}^n \frac{c_i}{a(t_i)}$. Notation: $C = \{(c_1, 0) \ldots, (c_n, n-1)\} = (c_1, c_2, \ldots, c_n)$

Principle of Equivalence When IR and $a(t)$ are the same, two cash flows are equivalent iff they have the same present value.

Time Value $TV_tC = PV(C) \times a(t)$, $PV(C) = \frac{TV_t(C)}{a(t)}$. For $0 < t_1 < t_2$, $TV_{t_2}(C) = \frac{a(t_2)}{a(t_1)} \times TV_{t_1}(C)$

Equation of Value For a cash flow $C = \{(c_1, t_1), (c_2, t_2), \ldots, (c_n, t_n)\}$, $PV(C) = \sum_{i=1}^n \frac{c_i}{(1+r)^{t_i}} = 0$

Law of One Price PV(what you pay) = PV(what you receive)

Newton Raphson Method To estimate non-negative root r of $PV(C) = 0$, a.k.a yield/**I**nternal **R**ate of **R**eturn

- First use Intermediate Value Theoreom to pin down a small interval containing the root
- Then pick an arbitrary point as the starting point α_0
- $\alpha_1 = \alpha_0 - \frac{f(\alpha_0)}{f'(\alpha_0)}$, $\alpha_{n+1} = \alpha_n - \frac{f(\alpha_n)}{f'(\alpha_n)}$

(Ex.) Find the IRR of $C = (-2, 1, 1, 1)$. Construct $PV(C) = 0$. Let $x = 1 + r$, set $f(x) = 0$, find $f'(x)$. Test values of $f(x)$ at $x = 1, 1.1, 1.2 \ldots$, observing where it crosses the y-axis. Use IVT to establish a range and set α_0 within the range.

Financial Instruments/Term Structure

Certificate of Deposit Time deposit, penalty for withdrawing early. Value after t years = $P \times a(t)$, earnings = $P \times (a(t) - 1)$. Interest rate earned for the kth year = $P \times (a(k) - a(k-1))$ (to calculate penalty)

Stock Eg. \$10 dividends at EOY, grow 3% per year. Mkt return is 15%. Expected stock price = $\sum_{t=1}^{\infty} \frac{10\times(1+3\%)^{t-1}}{a(t)}$, $a(t) = 1.15^t$

T-bills Short term debt by Govt. Issued at discount to par, no coupon, mature at par. $P(1+\lambda \times M) = R = F$ where M is years till maturity. Yield $\lambda = \frac{(F/P)-1}{M}$

Commercial Paper 1-2 mths debt by company. Issued at discount to par, no coupon, mature at par. Return is difference between maturity value and purchase price. Earnings = $P_1 - P_0$, rate of return = $\frac{P_1-P_0}{P_0}$, nominal rate of return = $\frac{P_1-P_0}{P_0} \times \frac{365}{t_2-t_1}$

Bond Valuation $P = \frac{R}{(1+\frac{\lambda}{m})^n} + \sum_{i=1}^n \frac{cF/m}{(1+\frac{\lambda}{m})^i}$

P = current price, F = face value (used by coupon rate), R = redemption value at maturity, c = nominal coupon rate, m = no. of coupon payments per year, n = total no. of coupon payments, λ = nominal yield

Bond yield is IRR of the bond at price P.

Assume $R = F$ unless otherwise stated. When $R = F$, $P = F + F\left(\frac{c-\lambda}{\lambda}\right)\left[1 - \frac{1}{(1+\frac{\lambda}{m})^n}\right]$

(i)$P > F$ iff $c > \lambda$ (premium), (ii)$P = F$ iff $c = \lambda$ (at par), (iii)$P < F$ iff $c < \lambda$ (at discount)

Price after kth payment $P_{k+1} = P_k(1+\frac{\lambda}{m}) - cF/m$

Zero Coupon Bond Price $P = \frac{R}{(1+\lambda)^n}$

Perpetual Bond Bond that never matures ($n \to \infty$) $P = \frac{cF}{\lambda}$

Taylor Series Approximate value of f near values of a. $f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^n(a)}{n!}(x-a)^n + \ldots$

Price resulting from change in yield

$P(\lambda + \Delta\lambda) = P(\lambda) + P'(\lambda) \cdot \Delta\lambda + \frac{P''(\lambda)}{2} \cdot (\Delta\lambda)^2 + \cdots + \frac{P^n(\lambda)}{n!} \cdot (\Delta\lambda)^n + \ldots$

Percentage change of bond price

$\frac{P(\lambda+\Delta\lambda)-P(\lambda)}{P(\lambda)} = \frac{P'(\lambda)}{P(\lambda)} \cdot \Delta\lambda + \frac{1}{2}\frac{P''(\lambda)}{P(\lambda)} \cdot (\Delta\lambda)^2 + \cdots + \frac{1}{n!}\frac{P^n(\lambda)}{P(\lambda)} \cdot (\Delta\lambda)^2 + \ldots$

Modified Duration $D_M = -\frac{P'(\lambda)}{P(\lambda)}$. Modified duration depends on type of yield.

Price Sensitivity Formula $\frac{P(\lambda+\Delta\lambda)-P(\lambda)}{P(\lambda)} \approx -D_M(\lambda) \cdot \Delta\lambda$. On the price-yield curve, $-D_M P$ is the slope of the tangent line at λ

Estimation of Price Change

New price $P(\lambda + \Delta\lambda) \approx P(\lambda) - (D_M P) \cdot \Delta\lambda$

Change in bond price $\Delta P = P(\lambda + \Delta\lambda) - P(\lambda) \approx -(D_M P) \cdot \Delta\lambda$

Notation For $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$, $tC = \{(t_i c_i, t_i) : i = 1, 2, \ldots, n\}$

For a bond with cash flow $C = \{(cF/M, i/m) : i = 1, 2, \ldots, n\} \cup \{(R, n/m)\}$, suppose λ is nominal yield compounded k times a year.

$$P(\lambda) = \frac{R}{(1+\lambda/k)^{kn/m}} + \sum_{i=1}^n \frac{cF/M}{(1+\lambda/k)^{ki/m}} = PV(C)$$

$$P'(\lambda) = -\frac{\frac{nR}{m}}{(1+\lambda/k)^{kn/m+1}} - \sum_{i=1}^n \frac{(\frac{i}{m})(\frac{cF}{m})}{(1+\frac{\lambda}{k})^{\frac{ki}{m}+1}}$$

$= \left(-\frac{1}{1+\lambda/k}\right) \cdot PV(tC) \Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{1}{1+\lambda/k} \cdot \frac{PV(tC)}{PV(C)}$.

When $k = 1$, $D_M(\lambda) = \frac{1}{1+\lambda} \cdot \frac{PV(tC)}{PV(C)}$ (i.e. λ is effective annual yield).

If λ is continuously compounded yield, $a(t) = \exp(\lambda t)$.

$$P(\lambda) = \frac{R}{\exp(\lambda(n/m))} + \sum_{i=1}^n \frac{cF/m}{exp(\lambda(i/m))} = PV(C)$$

$$P'(\lambda) = -\frac{nR/M}{\exp(\lambda(n/m))} - \sum_{i=1}^n \frac{cF/m}{\exp(\lambda(i/m))} = -PV(tC)$$

$\Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{PV(tC)}{PV(C)}$

(Macaulay) Duration: $D(\lambda)$ is the modified duration w.r.t the bond's continuously compounded yield. $MacD = D = \frac{PV(tC)}{PV(C)}$

If λ is nominal yield compounded k times per year, $D_M(\lambda) = \frac{1}{1+\lambda/k}D(\lambda)$. If λ is effective annual yield, $D_M(\lambda) = \frac{1}{1+\lambda}D(\lambda)$. If λ is continuously compounded yield, $D_M(\lambda) = D(\lambda)$. Note: Macaulay Duration does not depend on the type of yield.

Macaulay Duration for Bond Let $\mu = \frac{\lambda}{m}$, $\gamma = \frac{c}{m}$. Then $D(\lambda) = (1+\frac{\lambda}{m})D_M = \frac{1+\mu}{m\mu} - \frac{1+\mu+n\cdot(\gamma-\mu)}{m\gamma[(1+\mu)^n-1]+m\mu}$.

As $n \to \infty$, for a perpetual bond, $D = \frac{1+\mu}{m\mu} = \frac{1+\lambda/m}{\lambda}$. When bond is priced at par, $\mu = \gamma$, $D = \frac{1+\mu}{m\mu}\left(1 - \frac{1}{(1+\mu)^n}\right)$

Generalization to Other Cash Flows For $C = \{(c_i, t_i) : i = 1, 2, \ldots, n\}$.

$$P(\lambda) = \sum_{i=1}^n \frac{c_i}{a(t_i)} = \sum_{i=1}^n \frac{c_i}{(1+\frac{\lambda}{k})^{kt_i}} = PV(C)$$

$P'(\lambda) = -\frac{1}{1+\lambda/k} \cdot PV(tC) \Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{1}{1+\lambda/k} \cdot \frac{PV(tC)}{PV(C)}$. If $k = 1$ i.e. λ is effective annual interest rate, $D_M(\lambda) = \frac{1}{1+\lambda} \cdot \frac{PV(tC)}{PV(C)}$. If λ is continuously compounded interest rate, $a(t) = \exp(\lambda t)$, $P(\lambda) = PV(C)$, $P'(\lambda) = -PV(C) \Rightarrow D_M(\lambda) = -\frac{P'(\lambda)}{P(\lambda)} = \frac{PV(tC)}{PV(C)} \equiv D(\lambda)$. If λ is nominal yield compounded k times a year, $D_M(\lambda) = \frac{1}{1+\lambda/k}D(\lambda)$

Macaulay Duration (Time Weighted Average) $D = \frac{PV(tC)}{PV(C)} = \sum_{i=1}^n w_i t_i$, where $w_i = \frac{PV(c_i)}{\sum_{i=1}^n PV(c_i)}$. If $c_i \geq 0$ for all i, then $t_1 \leq D \leq t_n$. For a zero-coupon bond, $D = T$.

Convexity $C(\lambda) = \frac{P''(\lambda)}{P(\lambda)}$. 2nd order approximation of change in bond price for a change in yield. Consider the quadratic term in the Taylor series: $P(\lambda + \Delta\lambda) \approx P(\lambda) + P'(\lambda)\Delta\lambda + \frac{P''(\lambda)}{2}(\Delta\lambda)^2$. Note: C depends on the type of interest rate.

Percentage change in bond price $\frac{P(\lambda+\Delta\lambda)-P(\lambda)}{P(\lambda)} \approx \frac{P'(\lambda)}{P(\lambda)}\Delta\lambda + \frac{1}{2}\frac{P''(\lambda)}{P(\lambda)})(\Delta\lambda)^2 = -D_M\Delta\lambda + \frac{1}{2}C(\lambda)^2$.

$\Delta P = P(\lambda + \Delta\lambda) - P(\lambda) \approx P[-D_M\Delta\lambda + \frac{1}{2}C(\Delta\lambda)^2]$

If λ is nominal IR compounded k times a year, $P''(\lambda) = \frac{1}{(1+\lambda/k)^2}(PV(t^2C) + PV(tC)/k)$,

$C(\lambda) = \frac{P''(\lambda)}{P(\lambda)} = \frac{1}{(1+\lambda/k)^2} \cdot \left(\frac{PV(t^2C)}{PV(C)} + \frac{1}{k} \cdot \frac{PV(tC)}{PV(C)}\right)$

When $k = 1$ i.e. λ is effective annual IR, then $C(\lambda) = \frac{1}{(1+\lambda)^2} \cdot (\frac{PV(t^2C)+PV(tC)}{PV(C)})$

If λ is continuously compounded, $P(\lambda) = \sum_{i=1}^n c_i \exp(-t_i\lambda) = PV(C)$, $P'(\lambda) = -\sum_{i=1}^n c_i t_i \exp(-t_i\lambda) = -PV(tC)$, $P''(\lambda) = \sum_{i=1}^n c_i t_i^2 \exp(-t_i\lambda) = PV(t^2C)$. $\Rightarrow C(\lambda) = \frac{PV(t^2C)}{PV(C)}$

Macaulay Convexity $MacC = \frac{PV(t^2C)}{PV(C)}$. Macaulay convexity does not depend on the type of IR. It is the weighted sum of time squared.

If λ is nominal IR compounded k times a year, $C(\lambda) = \frac{1}{(1+\lambda/k)^2}(MacC(\lambda) + D(\lambda)/k)$.

If λ is effective annual IR, $C(\lambda) = \frac{1}{(1+\lambda)^2}(MacC(\lambda) + D(\lambda))$. If λ is continuously compounded IR, $C(\lambda) = MacC(\lambda)$.

Spot Rate Annual interest rate starting today ($t = 0$) until some time in the future, effectively the YTM of a ZCB maturing at t

$$(1+s_t)^t = (1+s_j)^j(1+f_{j,t})^{t-j}$$

$$(1+s_n)^n = (1+s_1)(1+f_{1,2})(1+f_{2,3})\ldots(1+f_{n-1,n})$$

Spot rate curve = spot rate against YTM

$P = \frac{F}{(1+s_n)^n} \Rightarrow s_n = (\frac{F}{P})^{1/n} - 1$

If λ is not known, use spot rate to find bond price.

Forward Rate Interest rate starting at $t_1 > 0$ lasting until $t_2 > t_1$, denoted f_{t_1,t_2}. Note: $f_{0,t} = s_t$. (Remember to subtract 1 from effective rate to get spot/forward rate)

Bootstrap Method Suppose $s_1, \ldots, s_{n-1}$ have already been determined. To estimate s_n, consider an n-period coupon paying bond with cash flow $(-P_n, cF, cF, \ldots, cF + F)$. Then $P_n = \frac{cF}{(1+s_1)} + \frac{cF}{(1+s_2)^2} + \cdots + \frac{cF}{(1+s_{n-1})^{n-1}} + \frac{cF+F}{(1+s_n)^n}$, so s_n can be determined.

Expected Utility Theory

Expected Utility $EU(X+w_0) = \sum_{i=1}^n p_i U(x_i + w_0)$ where $(x_1, \ldots, x_n)$ are n mutually exclusive payoffs with probabilities $(p_1, \ldots, p_n)$. For continuous X, $EU(X + w_0) = \int_a^b f(x)U(x + w_0)dx$

Utility-Based Decision Making Invest if $EU(X + w_0) > U(w_0)$, neutral if $=$, avoid if $<$. Between several risky prospects, just compare $EU(X_i + w_0)$ and take the highest

Risk attitude Risk averse iff U is strictly concave, neutral iff U is linear, loving if U is strictly convex

Concave (Strictly Concave) $U(\alpha x + (1-\alpha)y) \geq (>)\ \alpha U(x) + (1-\alpha)U(y)$

Convex (Strictly Convex) $U(\alpha x + (1-\alpha)y) \leq (<)\ \alpha U(x) + (1-\alpha)U(y)$.

Strictly convex if $U'' > 0$, linear if $U'' = 0$, strictly concave if $U'' < 0$

Jensen's Inequality If U is strictly concave, $E(U(X)) < U(E(X))$. Equality holds if U is linear, sign flipped if U is strictly convex

Fair Game zero expected payoff. $W = X + w_0$, $E(W) = w_0$. Risk averse individuals will always avoid a fair game since $E(U(W)) < U(E(W)) = U(w_0)$

Certainty equivalent the real number $c = CE(X; U)$ s.t. $U(c) = E(U(X + w_0))$. Invest if $c > w_0$, neutral if $=$, avoid if $<$

Risk premium the real number $r = RP(X; U)$ s.t. $U(w_0 - r) = E(U(w_0 + X))$. $r = w_0 - CE(X; U)$. In-

vest if $r < 0$, neutral if $=$, avoid if $>$

Positive Affine Transformations $CE(X; aX + b) = CE(X; U)$. $RP(X; U)$ is also constant under PAT

Absolute Risk Aversion (ARA) $-\frac{U''(w)}{U'(w)}$ at wealth level w. ARA> 0. It shows how an individual's risk aversion changes with wealth. $U_{ARA} = V_{ARA}$ iff U and V are positive affine transformations of each other (they are *equivalent*)

If $U_{ARA}(w) > V_{ARA}(w)$ for all w, U is *globally more risk averse* than V. Then there exists a strictly increasing and strictly concave function g s.t. $U(w) = g(V(w))$

Portfolio Selection $EU(W) = EU(w_0(1 + \alpha R))$. When R has two possible values a and b, $E(U(W)) = pU(w_0(1 + \alpha a)) + (1 - p)U(w_0(1 - \alpha b))$

Mean Variance Analysis

Rate of Return of Asset $r = \frac{P_1 - P_0}{P_0}$, mean $\mu_i = E(r_i)$, risk $= \sigma_i = \sqrt{Var(r_i)}$

RoR of Portfolio $r_p = \sum_{i=1}^n w_i r_i = \boldsymbol{w^T r}$

Correlation $\rho_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j} \leq 1$

Portfolio Mean $\mu_p = E(r_p) = \sum_{i=1}^n w_i \mu_i = \boldsymbol{w^T \mu}$ where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$

Portfolio Variance $\sigma_p^2 = Var(r_p) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \boldsymbol{w^T C w} = \sum_{i=1}^n w_i^2 \sigma_i^2 + \sum_{i=1}^n \sum_{j \neq i} w_i w_j \sigma_{ij}$

Portfolio of Two Assets $\boldsymbol{w} = (\alpha, 1 - \alpha)^T$, mean $\mu_p = \alpha\mu_1 + (1 - \alpha)\mu_2 \in [\mu_1, u_2]$, variance $\sigma_p^2 = \alpha^2\sigma_1^2 + (1 - \alpha)^2\sigma_2^2 + 2\alpha(1 - \alpha)\rho_{12}\sigma_1\sigma_2$

Global Minimum Portfolio Variance Occurs when $\alpha = \alpha^* = \frac{\sigma_2(\sigma_2 - \rho_{12}\sigma_1)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$, $(\sigma_p^2)^* = \frac{\sigma_1^2\sigma_2^2(1-\rho_{12}^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}$, $(\mu_p)^* = \alpha^*\mu_1 + (1 - \alpha^*)\mu_2$. GMVP $= (\sigma_g, \mu_g)$

Portfolio Graph $\sigma_p^2 = A\mu_p^2 + B\mu_p + C$ with $A > 0$

(Ex.) Given equation of min-variance frontier, find GMVP: $\sigma^2 = f(\mu) = \dots$, solve $f'(\mu) = 0$ or factorise into $x(\mu - y)^2$

(Ex.) Find the range which an investor would prefer p over q if he wants to minimize risk: Compare σ_p^2 against σ_q^2

(Ex.) Find α s.t. $\mu \geq k$: Method 1: $\mu = \alpha\mu_1 + (1 - \alpha)\mu_2 \geq k$, find range of α, find GMVP and compare with α^*. Method 2: substitute μ_p into equation of min-variance frontier, solve for GMVP

(Ex.) Find max μ s.t. $\sigma \leq k$: Substitute σ into the equation and choose the higher μ

Portfolio Theory

Lagrange Method Given $f(w_1, \dots, w_n)$ subject to $g_1(w_1, \dots, w_n) = 0$ and $g_2(w_1, \dots, w_n) = 0$, define $L(w_1, \dots, w_n, \lambda_1, \lambda_2) = f(w_1, \dots, w_n) - \lambda_1 g_1(w_1, \dots, w_n) - \lambda_2 g_2(w_1, \dots, w_n)$. $f(w_1, \dots, w_n)$ attains local extremum at w^* which is a solution to $\frac{\partial}{\partial w_i} L(w_1, \dots, w_n, \lambda_1, \lambda_2)$ for $i = 1, \dots, n$, $g_1 = 0$ and $g_2 = 0$

Find w that achieves μ with min variance $L = \frac{1}{2}\sigma_p^2 - \lambda_1\left(\sum_{i=1}^n w_i - 1\right) - \lambda_2\left(\sum_{i=1}^n w_i\mu_i - \mu\right) = \frac{1}{2}\boldsymbol{w^T C w} - \lambda_1(\boldsymbol{w^T 1} - 1) - \lambda_2(\boldsymbol{w^T \mu} - \mu) \implies \boldsymbol{Cw = \lambda_1 1 + \lambda_2 \mu}$, $\boldsymbol{1^T w} = 1$, $\boldsymbol{\mu^T w} = \mu$. Then $\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix}\lambda_1 \\ \lambda_2\end{pmatrix} = \begin{pmatrix}1 \\ \mu\end{pmatrix}$, $a = \boldsymbol{1^T C^{-1} 1}$, $b = \boldsymbol{1^T C^{-1} \mu}$, $c = \boldsymbol{\mu^T C^{-1} \mu}$, $\lambda_1 = \frac{c - b\mu}{ac - b^2}$, $\lambda_2 = \frac{a\mu - b}{ac - b^2}$

$$\boldsymbol{w_\mu^*} = \frac{a(c - b\mu)}{ac - b^2}\frac{\boldsymbol{C^{-1}1}}{\boldsymbol{1^T C^{-1} 1}} + \frac{b(a\mu - b)}{ac - b^2}\frac{\boldsymbol{C^{-1}\mu}}{\boldsymbol{1^T C^{-1}\mu}}$$

Variance $\sigma_\mu^2 = \boldsymbol{w_\mu^{*T} C w_\mu^*} = \frac{a\mu^2 - 2b\mu + c}{ac - b^2}$

GMVP $(\sigma_g, \mu_g) = (\sqrt{\frac{1}{a}}, \frac{b}{a})$, $\boldsymbol{w_{GMV}} = \frac{\boldsymbol{C^{-1}1}}{\boldsymbol{1^T C^{-1} 1}} = \frac{\boldsymbol{C^{-1}1}}{a}$

Min Variance Frontier $\sigma^2 = \frac{a(\mu - \frac{b}{a})^2 + (c - \frac{b}{a})^2}{ac - b^2}$ or $a\sigma^2 - \frac{a^2}{ac - b^2}(\mu - \frac{b}{a})^2 = 1$

Efficient Frontier The set of points $\mu_p \geq \mu_g = \frac{b}{a}$ (change to $\mu = f(\sigma)$)

Covariance of two rates $Cov(r_q, r_p) = \boldsymbol{w_p^T C w_q} = Cov(r_p, r_q)$, $Cov(r_g, r_q) = \frac{1}{\boldsymbol{1^T C^{-1} 1}} = \frac{1}{a} = \sigma_g^2$

Two Fund Theorem $\boldsymbol{w_\mu^*} = \alpha\frac{\boldsymbol{C^{-1}1}}{\boldsymbol{1^T C^{-1} 1}} + (1 - \alpha)\frac{\boldsymbol{C^{-1}\mu}}{\boldsymbol{1^T C^{-1}\mu}}$

Any portfolio lying on the min-variance frontier is a weighted sum of **any two** distinct min-variance portfolios $\boldsymbol{w_1}$ and $\boldsymbol{w_2}$. If $\mu = \alpha\mu_1 + (1 - \alpha)\mu_2$, then $\boldsymbol{w = \alpha w_1 + (1 - \alpha) w_2}$

Corollary If $b > 0$ then Fund 2 is efficient, and any portfolio with $\alpha < 1$ is efficient.

(Ex.) Find weight vector with smallest variance out of all portfolios equally weighted in asset 2 and 3:

Let $\boldsymbol{w} = (1 - 2\alpha, \alpha, \alpha)^T$, then $\sigma^2 = \boldsymbol{w^T C w} = f(\alpha)$. Solve for $f'(\alpha) = 0$.

(Ex.) Check if $\boldsymbol{w}$ is on efficient frontier: Given two other portfolios $\boldsymbol{w_{Fund1}}$ and $\boldsymbol{w_{Fund2}}$, check if there exists α s.t. $\boldsymbol{w = \alpha w_{Fund1} + (1 - \alpha) w_{Fund2}}$

Portfolio with Risk-Free Asset $(1 - \boldsymbol{1^T w}, w_1, \dots, w_n)^T$

Rate of Return $r_p = (1 - \boldsymbol{1^T w})r_f + \boldsymbol{w^T r}$ where $\boldsymbol{r} = (r_1, \dots, r_n)^T$, mean $\mu_p = (1 - \boldsymbol{1^T w})r_f + \boldsymbol{w^T \mu}$, variance $\sigma_p^2 = Var((1 - \boldsymbol{1^T w})r_f + \boldsymbol{w^T r}) = Var(\boldsymbol{w^T r}) = \boldsymbol{w^T C w}$

Find portfolio in n risky assets and 1 risk-free asset with min variance and $\mu_p = \mu$ Minimize $\boldsymbol{w^T C w}$ subject to $\mu_p = (1 - \boldsymbol{w^T 1})r_f + \boldsymbol{w^T \mu} = \mu$. $L = \frac{1}{2}\boldsymbol{w^T C w} - \lambda((1 - \boldsymbol{w^T 1})r_f + \boldsymbol{w^T \mu} - \mu) \implies \boldsymbol{Cw} = \lambda(\boldsymbol{\mu} - r_f\boldsymbol{1})$.

Then $\mu - r_f = \boldsymbol{\mu^T w} - r_f\boldsymbol{1^T w}$, $\lambda = \frac{\mu - r_f}{(\boldsymbol{\mu^T} - r_f\boldsymbol{1^T})\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}$

$$\boldsymbol{w} = \frac{(\boldsymbol{\mu} - r_f)\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}{(\boldsymbol{\mu} - r_f\boldsymbol{1})^T\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}$$

$(\boldsymbol{\mu} - r_f\boldsymbol{1})^T\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1}) = c - 2br_f + ar_f^2$

Variance $\sigma^2 = \boldsymbol{w^T C w} = \frac{(\mu - r_f)^2}{(\boldsymbol{\mu} - r_f\boldsymbol{1})^T\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})} = \frac{(\mu - r_f)^2}{c - 2br_f + ar_f^2}$, $\sigma = \frac{|\mu - r_f|}{\sqrt{c - 2br_f + ar_f^2}}$

It is assumed $\mu_{GMV} = \frac{b}{a} > r_f$

Tangency Portfolio The tangent point of the Capital Market Line (CML) with Min-Variance Frontier. It only invests in the n risky assets, $\sum_{i=1}^n w_i = 1$

$$\boldsymbol{w_{tan}} = \frac{\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}{\boldsymbol{1^T C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}$$

$$\mu_{tan} = \frac{\boldsymbol{\mu^T C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}{\boldsymbol{1^T C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})} = \frac{c - r_f b}{b - r_f a}$$

$\sigma_{tan}^2 = \frac{(\boldsymbol{\mu} - r_f\boldsymbol{1})^T\boldsymbol{C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1})}{(\boldsymbol{1^T C^{-1}}(\boldsymbol{\mu} - r_f\boldsymbol{1}))^2} = \frac{c - 2r_f b + ar_f^2}{(b - r_f a)^2}$

If $\mu_{GMV} = \frac{b}{a} < r_f$, tangency portfolio is below GMVP (lies on inefficient part of min-variance frontier).

If $\mu_{GMV} = \frac{b}{a} = r_f$, there is no tangency portfolio.

Equation of CML $\mu_p = r_f + \frac{\sigma_p}{\sigma_{tan}}(\mu_{tan} - r_f)$ where (σ_p, μ_p) is any point on the efficient frontier

Sharpe Ratio of portfolio p in $(n+1)$ assets $SR_p = \frac{\mu_p - r_f}{\sigma_p}$ (risk-adjusted return/expected excess return per unit risk) Portfolios that lie on the CML have highest Sharpe ratio.

(Ex.) Given equation of min-variance frontier and CML, find μ_{tan} and σ_{tan}^2: Substitute equation of CML into min-variance frontier, solve for either σ or μ

(Ex.) Find k if invested $k\%$ into risk-free asset and $(1 - k\%)$ in $\boldsymbol{w}_{tan}$:

$r_p = \frac{k}{100}r_f + (1 - \frac{k}{100})r_{tan}$, $\mu_p = E(r_p) = \frac{k}{100}r_f + (1 - \frac{k}{100})\mu_{tan}$

(Ex.) Given equation of min-variance frontier and r_f, find CML: Implicit differentiation of $\sigma^2 = A\mu^2 + B\mu + c$, find $\frac{d\mu}{d\sigma}$ and equate with gradient of CML $\frac{\mu_{tan} - r_f}{\sigma_{tan}}$.

CAPM

Mkt Portfolio Risky assets only $\boldsymbol{w} = (w_1, \dots, w_n)^T$, $w_i = n_i p_i / \sum n_i p_i$ i.e. $\boldsymbol{w}_{tan}$

Risk Premium $\mu_p - r_f$ (expected return in excess of the risk-free rate)

CAPM Theorem $\mu_i - r_f = \frac{\sigma_{im}}{\sigma_m^2}(\mu_m - r_f)$

Beta of Asset $\beta_i = \frac{\sigma_{im}}{\sigma_m^2}$

Portfolio Beta $\beta_p = \frac{\sigma_{pm}}{\sigma_m^2} = \frac{Cov((1 - \boldsymbol{1^T w})r_f + \sum w_i r_i, r_m)}{\sigma_m^2} = \frac{Cov(\sum w_i r_i, r_m)}{\sigma_m^2} = \frac{\sum w_i Cov(r_i, r_m)}{\sigma_m^2} = \boldsymbol{w^T \beta}$ where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_n)^T$

$$\boldsymbol{\beta} = \begin{pmatrix}\beta_1 \\ \vdots \\ \beta_n\end{pmatrix} = \begin{pmatrix}\sigma_{1m}/\sigma_m^2 \\ \vdots \\ \sigma_{nm}/\sigma_m^2\end{pmatrix} = \frac{1}{\sigma_m^2}\begin{pmatrix}\sigma_{1m} \\ \vdots \\ \sigma_{nm}\end{pmatrix} = \frac{1}{\sigma_m^2}\begin{pmatrix}\boldsymbol{e_1^T C w_m} \\ \vdots \\ \boldsymbol{e_n^T C w_m}\end{pmatrix} = \frac{1}{\sigma_m^2}\begin{pmatrix}\boldsymbol{e_1} \\ \vdots \\ \boldsymbol{e_n^T}\end{pmatrix}\boldsymbol{C w_m} = \frac{1}{\sigma_m^2}\boldsymbol{I C w_m} = \frac{1}{\sigma_m^2}\boldsymbol{C w_m}$$

$\implies \boldsymbol{1^T C^{-1}\beta} = \frac{1}{\sigma_m^2}\boldsymbol{1^T w_m} = \frac{1}{\sigma_m^2}$

$\implies \boldsymbol{\beta^T C^{-1}\beta} = \frac{1}{\sigma_m^2}\boldsymbol{\beta^T w_m} = \frac{1}{\sigma_m^2}\beta_m = \frac{1}{\sigma_m^2} = \boldsymbol{1^T C^{-1}\beta}$

$\implies \boldsymbol{w_m} = \frac{\boldsymbol{C^{-1}\beta}}{\boldsymbol{1^T C^{-1}\beta}}$

CAPM Theorem II $\mu_p - r_f = \beta_p(\mu_m - r_f)$. Note that $\beta_m = 1$.

Security Market Line $\mu_i = r_f + \beta_i(\mu_m - r_f)$

Mispricing occurs when risk premium does not fall on SML. If $\mu_i <$ SML, asset is overpriced. If $\mu_i >$ SML, underpriced

Alpha Diff btw estimated return and the benchmark return is known as the *alpha* of the return. Underpriced assets have positive alphas while overpriced assets have negative alphas

Systematic and Non-systematic risk

$r_p = r_f + \beta_p(r_m - r_f) + \epsilon_p$. $E(\epsilon_p) = 0$, $Cov(\epsilon_p, r_m) = 0$. $Var(r_p) = \beta_p^2\sigma_m^2 + Var(\epsilon_p)$

Total risk of any portfolio is a sum of two components: the systematic risk $\beta_p^2\sigma_m^2$ and non-systematic (idiosyncratic) risk $Var(\epsilon_p)$. Systematic risk cannot be diversified away but non-systematic risk is uncorrelated with the market and can be reduced through diversification

Portfolios on CML A portfolio contains only systematic risk iff it is on the CML. $r_p = \alpha r_m + (1 - \alpha)r_f$, $\sigma_p^2 = \beta_p^2\sigma_m^2$. For a portfolio A on the right of the CML, there is an efficient portfolio E s.t. $\sigma_A^2 > \sigma_E^2 = \beta^2\sigma_m^2$ (A carries non-systematic risk)

(Ex.) A portfolio invests 50% into risk-free asset, α into A and the rest into B, and $\beta_p = k$, find α and μ_p: $\beta_p = \boldsymbol{w^T \beta}$, solve for α. Then $\mu_p = 0.5r_f + \alpha\mu_A + (0.5 - \alpha)\mu_B$. Note that $\beta_f = 0$

(Ex.) Find the variance of $r_p = \frac{3}{5}r_A + \frac{2}{5}r_B$: $\sigma_m^2 = Var(r_p) = \frac{9}{25}\sigma_A^2 + 2\rho(\frac{3}{5})(\frac{2}{5})\sigma_A\sigma_B + \frac{4}{25}\sigma_B^2$

Covariance Properties $Cov(X, a) = 0$, $Cov(X, X) = Var(X)$, $Cov(aX, bY) = abCov(X, Y)$, $Cov(a + X, b + Y) = Cov(X, Y)$, $Var(\sum a_i X_i) = \sum_{ij} a_i a_j Cov(X_i, X_j)$

$Cov(r_A, r_B) = \rho_{AB}\sigma_A\sigma_B$

If two variables are uncorrelated, $Cov(X, Y) = 0$

Bilinearity of Covariance $Cov(X, aY + bZ) = aCov(X, Y) + bCov(X, Z)$

Variance Properties $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2ab \cdot Cov(X, Y)$

$Var(X + Y) = Var(X) + Var(Y) + 2 \cdot Cov(X, Y)$

$Var(cX) = c^2Var(X)$, $Var(c + X) = Var(X)$, $Var(c) = 0$